1 Preliminary Numerical Study of Gray-Cat Metastable Interfaces, Weak C Readout, and Strong D Observation Selection in a Closed Phase System v1

Date: 2026-07-14
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Series: Wave Information Readout / preliminary numerical study of A/B allocation metastability and observation selection in a closed system
Version DOI: 10.5281/zenodo.21353209
Concept DOI: 10.5281/zenodo.21353208

1.1 0. Conclusion

This paper summarizes a preliminary numerical experiment in which the white-cat state A and the black-cat state B are represented by closed-system complex amplitudes

$$a, \quad b$$

and read through

$$p_A = |a|^2, \qquad p_B = |b|^2,$$

$$Q = p_A + p_B, \qquad S = p_A - p_B.$$

The white-cat, black-cat, and gray-cat terminology is used here only as a metaphor for A/B allocation states. It is not a claim to reproduce the standard Schrodinger-cat thought experiment itself.

The experiment confirms the following points.

```
1. In the AB two-body system alone, a gray eigen phase,
   a gray metastable phase, and a natural selection phase were separated.
2. Under weak C readout, a window existed in which the gray metastable
   phase could be read without selecting one side.
3. Under strong D observation, the gray metastable phase was selected
   into either the white-cat or black-cat allocation.
4. In the gray eigen phase, the gray allocation was kept even under
   strong D observation.
5. No-D controls did not show equivalent white/black selection, so the
   selection could be separated as D-induced in the tested cases.
```

Thus the observed classification is:

```
gray metastable phase:
  weak readout reads it as gray;
  strong observation selects white or black.

gray eigen phase:
  weak readout reads it as gray;
  strong observation keeps it gray.
```

When the result is white-selected,

$$S \simeq +1, \qquad p_A \simeq 1, \qquad p_B \simeq 0.$$

When the result is black-selected,

$$S \simeq -1, \qquad p_A \simeq 0, \qquad p_B \simeq 1.$$

This should be read as an almost one-sided relocation of the A/B allocation, not as the appearance of two cats. The closed quantity

$$Q = p_A + p_B$$

is preserved.

1.2 1. Background and Aim

The wave-information-readout series has studied whether position-like, momentum-like, energy-like, and acceleration-like quantities can be constructed from closed complex phase relations and interference readouts without assuming a background space in advance.

The preceding acceleration-basis and localization-exchange experiment studied how localization and effective harmonic order are redistributed by an AB interaction based on an exchange-interference scattering matrix.

This paper moves that line of work to the selection of an A/B allocation state.

The question is:

```
Is a mixed white-cat A and black-cat B allocation merely an unselected state,
or can it be held as a gray-cat eigen state?

How does the state respond to weak C readout and strong D observation?
```

The experiment is divided into four stages.

```
Stage 1:
  Search for gray eigen, gray metastable, and natural selection phases
  in the AB two-body system alone.

Stage 2:
  Test whether weak C readout can read the gray metastable phase without
  selecting one side.

Stage 3:
  Test whether strong D observation selects the gray metastable phase
  into white or black.

Stage 4:
  Sweep the D observation start step and D gain to read the selection boundary.
```

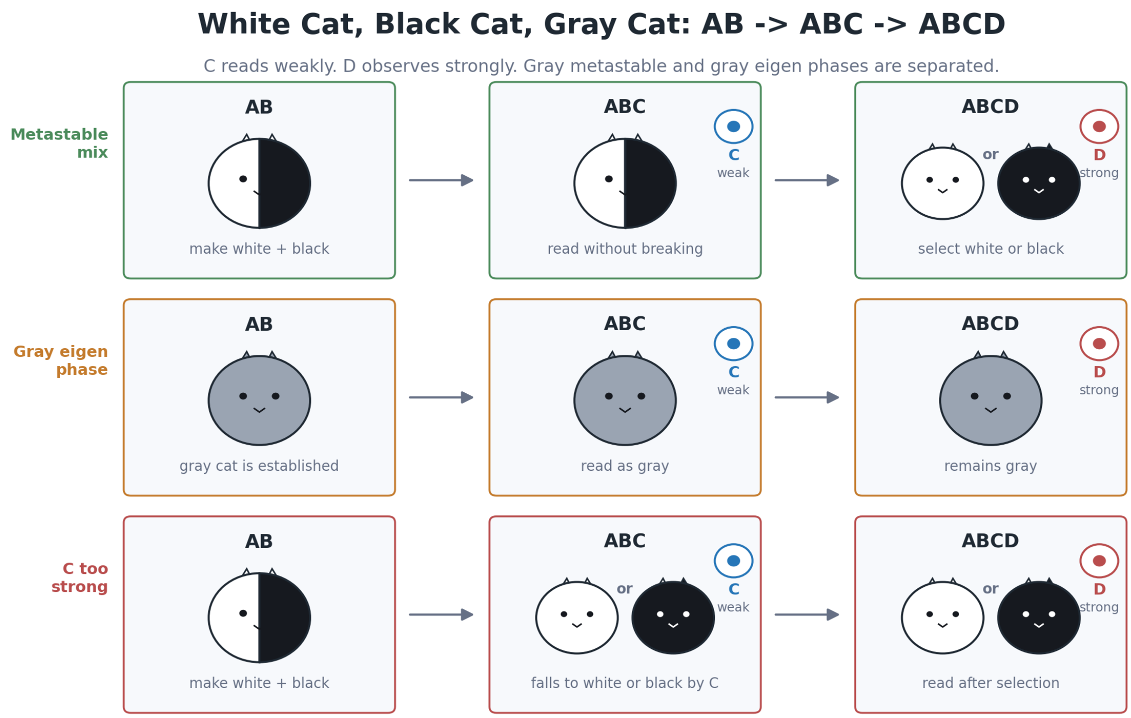

Figure 1: White, black, and gray cat AB-ABC-ABCD state transition diagram

1.3 1.1 State Transition Diagram

The three representative paths are shown below along the AB, ABC, and ABCD stages.

The upper path shows a metastable white+black mixture prepared in AB, weakly read by C, and selected into white or black by strong D observation.

The middle path shows a gray eigen phase already formed at the AB stage and kept gray under both weak C readout and strong D observation.

The lower path shows the case in which C is too strong and selection into white or black already occurs at the ABC stage.

1.4 1.2 Observed-Value Transition Diagram

The same three paths are also shown using the computed values of `p_A`, `p_B`, and `S`.

The solid curves are the internal values `p_A` and `p_B`. The dashed curve is the selection order variable mapped to the same vertical axis as

$$\frac{S+1}{2}.$$

The dotted curves represent the C readout in the ABC interval and the D readout in the ABCD interval.

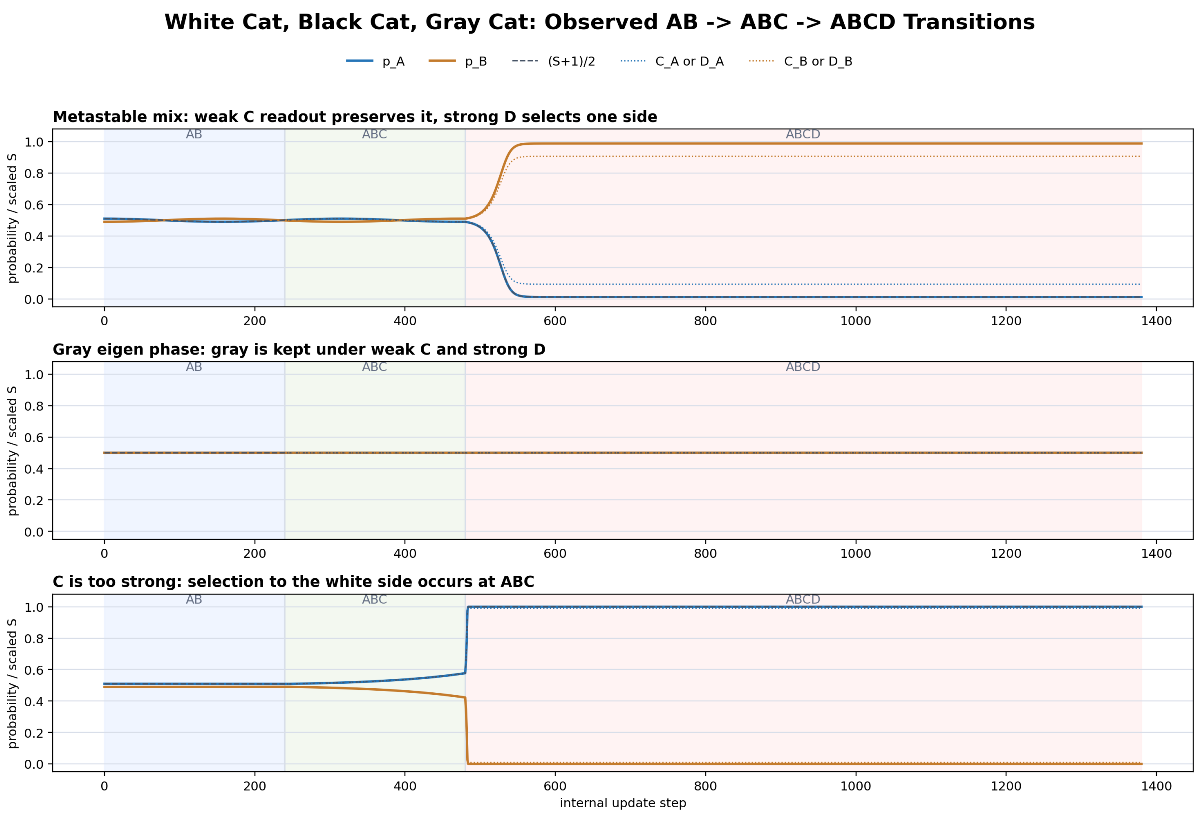

Figure 2: Observed-value transition of white, black, and gray cat states

1.5 2. Variables and Phase Classification

1.6 2.1 A/B Allocation

The A/B allocation is read as

$$p_A = |a|^2, \qquad p_B = |b|^2.$$

The closed quantity is

$$Q = p_A + p_B.$$

The A/B selection order variable is

$$S = p_A - p_B.$$

Then

$$p_A = \frac{1+S}{2}, \qquad p_B = \frac{1-S}{2}.$$

1.7 2.2 White, Black, and Gray Cat Readouts

State	Numerical guide	Readout
white cat	`S≈+1`	`p_A≈1`, `p_B≈0`
black cat	`S≈-1`	`p_A≈0`, `p_B≈1`
gray cat	`S≈0`	`p_A≈0.5`, `p_B≈0.5`

1.8 2.3 Phase Classification

For the AB-only evolution, the phases are classified as follows.

```
gray_eigen:
  S_mean ≈ 0
  S_amp ≈ 0
  S_drift ≈ 0

gray_metastable:
  S_mean ≈ 0
  0 < S_amp < S_gray_limit
  S_drift ≈ 0

natural_selection:
  S -> +1 or S -> -1 even without C or D.

large_oscillation:
  S oscillates strongly and a clear A/B bias appears.
```

The main targets of this paper are `gray_metastable` and `gray_eigen`.

1.9 3. Model

1.10 3.1 AB Exchange Interaction

The minimal AB exchange map is

$$\begin{pmatrix} a_{k+1} \\ b_{k+1} \end{pmatrix} = U_\epsilon \begin{pmatrix} a_k \\ b_k \end{pmatrix},$$

$$U_\epsilon = \begin{pmatrix} \cos\epsilon & i\sin\epsilon \\ i\sin\epsilon & \cos\epsilon \end{pmatrix}.$$

A weak restoring term or a weak nonlinear term may be added to test closed-system stability. However, conditions that naturally select one side without C or D are not counted as observation-induced selection.

1.11 3.2 Weak C Readout

C is an internal readout wave that weakly reads the A/B allocation. The visibility is

$$v_C = \frac{g_C}{g_C + \kappa_C}.$$

Then

$$S_C = v_C S,$$

$$C_A = \frac{1 + S_C}{2}, \qquad C_B = \frac{1 - S_C}{2}.$$

C is used to search for a window where the A/B allocation can be read without selecting one side.

1.12 3.3 Strong D Observation

D is a strong observation map that amplifies the currently read direction of `S`. Its visibility is

$$v_D = \frac{g_D}{g_D + \kappa_D},$$

and

$$S_D = v_D S.$$

The D backaction is implemented as

$$S_{k+1} = S_k + G_D S_D (1 - S_k^2),$$

where `G_D` is the D gain.

For the same D start state, a no-D control is run in parallel. If the no-D control also falls into the same white-cat or black-cat selection, the result is not counted as D-induced selection.

1.13 4. Stage 1: AB Metastable Interface Search

1.14 4.1 Conditions

```
C = off
D = off
steps = 4096
S_gray_limit = 0.05
selection_limit = 0.95
```

The main swept parameters were:

```
epsilon_values = (0.0, 1e-05, 3e-05, 0.0001, 0.0003, 0.001, 0.003, 0.01)
stability_gain_values = (-0.01, -0.002, 0.0, 0.002, 0.01)
```

1.15 4.2 Results

phase	count
gray_eigen	1248
gray_metastable	733
large_oscillation	470
natural_selection	1070
unstable_or_drifting	2079

In the AB system alone, the gray eigen phase, gray metastable phase, large-oscillation region, and natural selection phase were separated.

Representative gray-metastable candidates were:

epsilon	phi/pi	s0	gain	S_mean	S_amp	S_drift
0.01	0	0.01	0	0.000173315	0.02	0.00205793
0.01	1	0.01	0	0.000173315	0.02	0.00205793
0.003	0.0833333	0	-0.002	-0.00213507	0.020193	0.00227521

1.16 5. Stage 2: Weak C Readout Window

1.17 5.1 Conditions

```
C = on
D = off
```

```
steps = 4096
readout_kappa = 0.02
g_C_values = (0.0, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1.0)
backaction_scale_values = (0.0, 1e-05, 5e-05, 0.0001, 0.0005, 0.001, 0.002, 0.005, 0.01)
```

1.18 5.2 Results

```
total_cases = 1080
C_window_count = 247
C_informative_window_count = 144
C_nonzero_backaction_window_count = 114
```

The phase classification after C was:

phase_after_C	count
gray_eigen	90
gray_metastable	786
large_oscillation	141
natural_selection	2
unstable_or_drifting	61

For gray-metastable candidates, windows were found in which C could read the A/B allocation without selecting one side.

Representative nonzero-backaction C windows were:

epsilon	phi/pi	s0	base_gain	g_C	c_gain	C_rel_err	C_bias_delta	phase_after_C
0.003	0.0833333	0	-0.002	1	5e-05	0.0196078	0.00122196	gray_metastable
0.003	0.0833333	0	-0.002	1	1e-05	0.0196078	0.000220996	gray_metastable
0.003	0.0833333	0.001	-0.002	1	1e-05	0.0196078	0.000222243	gray_metastable

---

1.19 6. Stage 3: Strong D Observation Response

1.20 6.1 Conditions

```
Stage 1: AB metastable interface already searched
Stage 2: C readout window already confirmed
Stage 3: D strong observation response
pre_steps_values = (0, 1, 2, 5, 10, 20, 50, 100, 250, 500, 1000, 2000)
d_steps = 2048
c_modes = (record_only, weak_C_window)
g_D_values = (0.0, 0.05, 0.1, 0.2, 0.5, 1.0)
d_backaction_scale_values = (0.0, 0.02, 0.05, 0.1, 0.2, 0.5, 1.0)
```

1.21 6.2 Overall Results

```
total_cases = 7056
D_induced_selection_count = 2016
```

The D outcome classification was:

D_outcome	count
white_selected	1244
black_selected	772
gray_kept_eigen	1062
unresolved	3978

The no-D control was:

baseline_outcome	count
gray_kept_eigen	1176
unresolved	5880

The no-D control did not detect equivalent falling into white-cat or black-cat selection.

1.22 6.3 Gray Eigen Phase

In the gray eigen phase, the following was confirmed even under strong D conditions.

pre	C_mode	S_start	outcome	S_mean_after_D	S_amp_after_D	Q_err
0	record_only	0	gray_kept_eigen	0	0	2.22e-16
0	weak_C_window	0	gray_kept_eigen	0	0	2.22e-16
20	record_only	0	gray_kept_eigen	0	0	2.22e-16
20	weak_C_window	0	gray_kept_eigen	0	0	2.22e-16

The gray eigen phase did not fall into white-cat or black-cat selection under strong D observation.

1.23 6.4 Gray Metastable Phase

In the gray metastable phase, strong D observation selected the state into white or black.

Representative cases were:

epsilon	phi/pi	s0	gain	pre	S_start	g_D	D_gain	outcome	baseline	S_mean_after_D
0.003	0.0833333	0	-0.002	0	0	1	1	black_selected	unresolved	-1
0.003	0.0833333	0	-0.002	0	0	0.5	0.25	black_selected	unresolved	-0.999878
0.01	0	0.01	0	0	0.02	1	0.2	white_selected	unresolved	0.997467

1.24 6.5 Large-Amplitude Separated Region

In the large-amplitude separated region, D outcomes agreed with the sign of the C readout.

pre	S_start	C_sign	outcome	S_mean_after_D	agreement
0	0.06	A	white_selected	1	same_sign

pre	S_start	C_sign	outcome	S_mean_after_D	agreement
1	0.0556528	A	white_selected	1	same_sign
2	0.0513121	A	white_selected	1	same_sign

In this region, D is interpreted as strongly reading an A/B bias already present in the system, rather than newly creating selection.

1.25 7. Stage 4: D Selection Boundary

1.26 7.1 Conditions

The targets were gray-metastable candidates for which D-induced selection was confirmed in Stage 3.

```
target = gray_metastable candidates only
d_steps = 2048
pre_steps_values_count = 73
C_modes = (record_only, weak_C_window)
```

The D gain was scanned finely near the boundary.

```
D_gain_values =
(0.0, 0.005, 0.01, 0.015, 0.02,
 0.0225, 0.025, 0.0275, 0.03,
 0.0325, 0.035, 0.0375, 0.04,
 0.045, 0.05, 0.055, 0.06,
 0.065, 0.07, 0.0725, 0.075,
 0.0775, 0.08, 0.0825, 0.085,
 0.0875, 0.09, 0.095, 0.1,
 0.12, 0.15, 0.2, 0.3,
 0.5, 0.75, 1.0)
```

1.27 7.2 Results

```
total_rows = 15768
boundary_points = 438
selection_possible_boundary_points = 438
no_selection_boundary_points = 0
min_D_gain_overall = 0.0225
max_min_D_gain_overall = 0.065
```

The boundary by candidate was:

case_id	boundary_points	selection_possible	no_selection	min_gain	max_min_gain	sign_counts
gray_metastable_0_eps0.01_phi0_s0.01_g0	146	146	0	0.065	0.065	A:90, B:56

case_id	boundary_points	selection_possible	no_selection	min_gain	max_min_gain	sign_counts
`gray_metastable_1_eps0.01_phi_s0.01_g0`	146	146	0	0.065	0.065	A:90, B:56
`gray_metastable_4_eps0.003_phi0.0833333_s0_g-0.002`	146	146	0	0.0225	0.0225	B:114, A:32

Within the swept range, D-induced selection was possible at all tested observation start steps for the selected metastable candidates.

The minimum D gain separated into two levels:

```
weak threshold candidate:
  min_D_gain = 0.0225

strong threshold candidates:
  min_D_gain = 0.065
```

1.28 8. Discussion

The important point in this experiment is that the gray-cat state was not a single type of state.

The gray metastable phase can be read as gray by weak readout. Under strong D observation, it falls into either white or black.

If it becomes white,

$$p_A \simeq 1, \qquad p_B \simeq 0.$$

If it becomes black,

$$p_A \simeq 0, \qquad p_B \simeq 1.$$

This is not the simultaneous appearance of a white cat and a black cat. It is an A/B allocation moving almost entirely to one side, with the opposite component almost disappearing in the readout.

In contrast, the gray eigen phase kept

$$p_A \simeq 0.5, \qquad p_B \simeq 0.5$$

even under strong D observation.

Thus the gray metastable phase and the gray eigen phase are distinguishable by their observation response.

Weak C readout also showed a window in which A/B allocation could be read without selection.

The experiment therefore separates:

```
natural selection in AB alone;
gray metastability readable by weak C readout without selection;
white/black selection under strong D observation.
```

1.29 9. Conclusion

This experiment confirmed the following points for an A/B allocation state in a closed system.

```
1. In AB alone, gray eigen, gray metastable, and natural selection phases
   can be classified.
2. Weak C readout has windows that read a gray metastable phase without
   selecting one side.
3. Strong D observation selects a gray metastable phase into white or black.
4. A gray eigen phase remains gray even under strong D observation.
5. In the D selection boundary sweep, all selected metastable candidates
   admitted D-induced selection within the tested range.
6. The minimum D gain separated into two levels, 0.0225 and 0.065,
   depending on the candidate.
```

The white-cat, black-cat, and gray-cat states are therefore not treated here as a single unselected state. They are separated by closed-system phase classification and observation strength.

In particular:

```
gray metastable phase:
  gray under weak readout;
  white or black under strong observation.

gray eigen phase:
  gray even under strong observation.
```